

robustness_summary_table

Check	Result
Repository-clustered SE	Main stage-specific interpretation remains qualitatively stable.
Exclude largest/top-5 repositories	Gate 0 and Gate 1 are stable; Gate 2 is more sensitive when PA-only sample size shrinks.
Exclude dominant/top-2 languages	Gate 0 and Gate 1 are stable; Gate 2 Context may weaken under reduced PA-only samples.
Exclude top 1%/5% PR sizes	Main effects remain qualitatively stable and are not driven by extreme PR-size outliers.
Alternative Gate 2 reuse levels	Context remains the strongest substantive Gate 2 signal, though significance varies by operationalization.
PQS-only models	PQS is informative for Gate 0 and Gate 1 but is less diagnostic than individual dimensions for Gate 2.
Prompt-length controls	Stage-specific interpretation remains intact after controlling for approximate prompt length.
Stratified 80/20 hold-out	Coefficient directions are generally consistent with the full-sample interpretation.